

Drawings and PCB Photos - Automated Part Costing

Problem Statement

PII:

No

- Costing a circuit board means listing every component on it and pricing each one. A medium-complexity board takes two to three weeks.
- It needs someone who knows what electronics cost — a small capacitor, a six-layer board, placing a fine-pitch chip. Few people have that, so boards wait for them.
- On mechanical parts, most of what we need is written on the 2D drawing, not held in the 3D model, and is read by eye and typed in by hand.
- The costing method and the rate data do not change. Only the reading changes.

Projected ROI

Cost: £???

Return: £???

What is the return based on?

The boards we turn away, and the weeks a specialist spends on each one.

Time (hrs/week)

2-3 weeks per board

Current State

- Costing one board means listing every component, pricing each line, then estimating the bare board and the assembly. Two to three weeks for a medium board.
- It needs an electronics cost specialist, and a design change means starting again.
- For a machined part, the drawing is read by eye and typed into CAPPe by hand.

Proposed Solution/Model

- Conduct a 3-stage Proof of Value (PoV), run the same way as the CAD to Cost PoV.
- Upload a photo of the board, or a drawing alongside the 3D model. The tool reads it and returns a full cost.
- Everything it reads is checked before it is used. It does not set any price.
- The detail is on the next page.

Expected Benefits:

- **No Specialist Needed:** A cost engineer can cost a board without being an electronics specialist.
- **Wider Coverage:** We can cost the boards and parts we turn away today.

Technical Scoping:

Enabling AI on an existing system?
(YES)

The reading is done inside CostVision. CAPPe is not changed.

Model used: Claude (Anthropic)

Sonnet and Haiku to read, Opus where a board needs a closer look.

Technical implementation/support required?
(YES)

It needs an approved route to the model. Option 2 runs with that route switched off. This one cannot.

Drawings and PCB Photos - The Proposal in Detail

Proposal

How a board is costed today

- Someone identifies every component from a photo or a sample — package, value, rating, and the exact part number for each chip.
- They find a price for every line, at the volume we are asking about.
- The bare board is estimated separately for size, layers, finish and test.
- Then the assembly: how many placements, how many are fine-pitch or BGA, one side or two.
- Two to three weeks for a medium board, and only a specialist can do it.

What we propose

- Extend the Proof of Value to cover 2D drawings and photographs of boards.
- Upload a photo of the board. Say how many you are making and where it is built.
- The tool lists the components it can see, prices each line from the catalogue, works out the bare board and the assembly, and returns the full cost breakdown.
- For a machined part, the drawing is read alongside the 3D model, for the tolerances, finish and heat treatment the model cannot carry.
- Every figure it reads is checked against the price catalogue or the measured geometry before it is used.
- Compare the output against cost estimates we already have.

Expected Benefits:

- A cost engineer can cost a board without being an electronics pricing specialist. The specialist checks the answer instead of building it.
- We can cost the boards and the parts we turn away today, because a photograph or a drawing is all we have.
- A cost can be redone when the design changes, instead of going out of date and staying that way.
- The costing method and the rate data do not change, so the answer is consistent with what we produce today and with everything downstream.
- Boards can be costed early, while the design can still be changed cheaply.
- Cost engineers stay in charge. Every figure is theirs to check, correct and sign off, and the tool never sets a price itself.